

In[1543]:=

```
ClearAll["Global`*"];
|清除全部

SeedRandom[];
|随机种子

(* ===== *)
(*1. 并发暴胀引擎 (完全原生 Union 去重, 无人工排序干预)*)
|并集

(* ===== *)
StepEvolution[g_Graph, window_] := Module[
|图 |模块
{allEdges, activeEdges, nActive, usedEdges, updates, tempE1, tempE2, x, y, z, wCounter,
newVertices, newActive, newFrozen, remainingEdges}, allEdges = EdgeList[g];
|边的列表

activeEdges = RandomSample[Cases[allEdges, _UndirectedEdge]];
|伪随机采样 |模式匹配 |无向边

nActive = Length[activeEdges];
|长度

If[nActive < 2, Return[{g, 0}]];
|如果 |返回

usedEdges = {}; updates = {};
Do[tempE1 = activeEdges[[i]];
|Do循环

If[! MemberQ[usedEdges, tempE1], Do[tempE2 = activeEdges[[j]];
|如果 |成员判定 |Do循环

If[! MemberQ[usedEdges, tempE2] &&
|如果 |成员判定

Length[Intersection[List @@ tempE1, List @@ tempE2]] == 1,
|长度 |交集 |列表 |列表

AppendTo[updates, {tempE1, tempE2}];
|附加

AppendTo[usedEdges, tempE1]; AppendTo[usedEdges, tempE2];
|附加 |附加

Break[];], {j, i + 1, Min[i + window, nActive]}]], {i, 1, nActive}];
|跳出循环 |最小值

If[Length[updates] == 0, Return[{g, 0}]];
|... |长度 |返回

wCounter = Max[VertexList[g]];
|... |顶点列表

newVertices = {}; newActive = {}; newFrozen = {};
Do[y = Intersection[List @@ pair[[1], List @@ pair[[2]]][1];
|Do循环 |交集 |列表 |列表

x = Complement[List @@ pair[[1], {y}]] [1];
|补集 |列表

z = Complement[List @@ pair[[2], {y}]] [1];
|补集 |列表

wCounter++;
AppendTo[newVertices, wCounter];
|附加

newActive = Join[newActive, {UndirectedEdge[x, z],
|连接 |无向边

UndirectedEdge[x, wCounter], UndirectedEdge[wCounter, z]}];
|无向边 |无向边
```

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newFrozen = Join[newFrozen, {DirectedEdge[x, y], DirectedEdge[y, z]}];
      连接      有向边      有向边
{pair, updates}];
remainingEdges = Complement[allEdges, Flatten[updates]];
      补集      压平
{Graph[Union[Join[VertexList[g], newVertices]],
  图      并集      连接      顶点列表
    Union[Join[remainingEdges, newActive, newFrozen]]], Length[updates]]];
      并集      连接      长度

(* =====*)
(*2. 并发宇宙演化采集 (40 步)*)
(* =====*)
Print[Style["[1/2] 正在点燃并发暴胀宇宙...", Blue, Bold]];
打印      样式      蓝色      粗体
universe = CompleteGraph[6];
      完全图
steps = 100;
widthData = {};

Monitor[Do[{universe, concurrentCount} = StepEvolution[universe, 50];
  监控      Do循环
    If[concurrentCount > 0,
      如果
        activeNodes = Union[Flatten[List@@Cases[EdgeList[universe], _UndirectedEdge]];
          并集      压平      列表      模...      边的列表      无向边
        If[Length[activeNodes] > 1, dists = (GraphDistance[universe, 1, #] & /@ activeNodes);
          ...      长度      图中顶点的距离
        AppendTo[widthData, {i, StandardDeviation[Select[dists, NumericQ]]}];];];
          附加      标准偏差      选择      数值量判定
        {i, 1, steps}], ProgressIndicator[i, {0, steps}]];
          进度指示器

(* =====*)
(*3. TCCT 全息肖尔分析 (Topological Sholl Analysis)*)
(* =====*)
Print[Style["\n[2/2] 演化停止, 启动全息肖尔形态学分析...", Purple, Bold, 14]];
打印      样式      紫色      粗体

(*强制将当前宇宙退化为 4D 纯因果连通图*)
      偏导
undirectedUniverse =
  Graph[VertexList[universe], UndirectedEdge@@@EdgeList[universe]];
      图      顶点列表      无向边      边的列表
seedNode = 1; (*1 号节点作为大爆炸原初奇点*)

(*发射全局测地线*)
distances = GraphDistance[undirectedUniverse, seedNode];
      图中顶点的距离
shellCounts = Counts[distances];
      关联计数
shollData = Sort@KeyValueMap[List, shellCounts];
      排序      键值映射      列表

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radii = shollData[[All, 1]];
volumes = shollData[[All, 2]];

(*提取核心拓扑参量*)
shollCriticalValue = Max[volumes];
shollCriticalRadius = radii[[First[Position[volumes, shollCriticalValue]]][1]];
primaryBranches = VertexDegree[undirectedUniverse, seedNode];
ramificationIndex = N[shollCriticalValue / primaryBranches];

decayData = Select[shollData, #[[1]] > shollCriticalRadius &];
If[Length[decayData] > 2, logDecayData = {#[[1]], Log[#[[2]]]} & /@ decayData;
fitDecay = LinearModelFit[logDecayData, x, x];
shollDecayK = fitDecay["BestFitParameters"][[2]];
r2Decay = fitDecay["RSquared"]; shollDecayK = 0; r2Decay = 0;];

(*绘制:肖尔宇宙剖面图*)
shollPlot = Show[ListLinePlot[shollData, PlotStyle → Directive[Thick, Darker[Cyan]],
Filling → Axis, FillingStyle → Opacity[0.2, Cyan], PlotTheme → "Detailed",
ImageSize → 600, FrameLabel → {Style["Topological Radius (r)", 12, Bold],
Style["Spatial Volume V(r)", 12, Bold]},
PlotLabel → Style["Cosmological Sholl Profile", 14, Bold], Graphics[
{Red, PointSize[0.03], Point[{0, shollData[[1, 2]]}], {Purple, Dashed, Thick, Line[
{shollCriticalRadius, 0}, {shollCriticalRadius, shollCriticalValue}]}]}];

Print[shollPlot];

(*输出物理张量表*)
shollTable = {{"Sholl Metric (Neuro-Morphology)",
"Cosmological Physical Mapping", "Extracted Value"},
{"Primary Branches", "Big Bang Initial Degrees of Freedom", primaryBranches},
{"Sholl Critical Radius (r_c)", "Turnaround Time (Expansion to Collapse)",
shollCriticalRadius}, {"Sholl Critical Value (N_max)",
"Maximum Equator Spatial Volume", shollCriticalValue}, {"Ramification Index",
"Inflation Amplification Factor", NumberForm[ramificationIndex, {Infinity, 2}]},
{"Sholl Regression Coeff (k)", "Big Crunch Topological Deceleration",

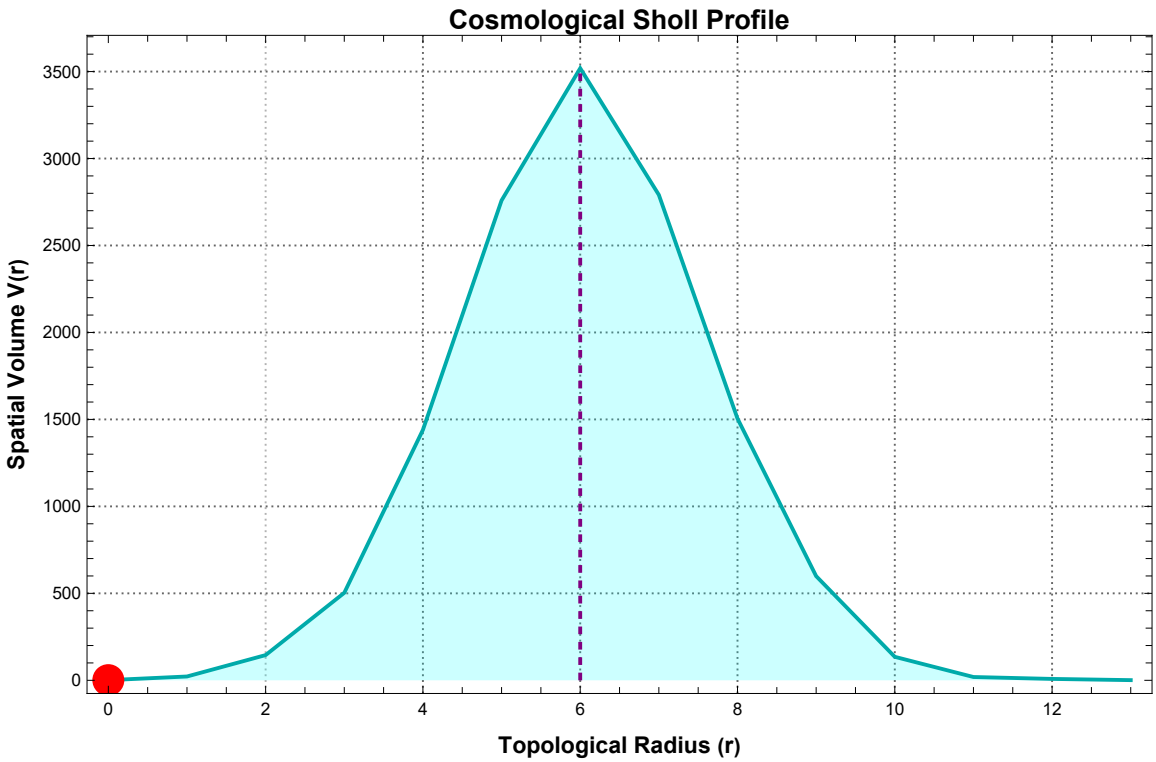
```

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If[shollDecayK ≠ 0, ToString[NumberForm[shollDecayK, {Infinity, 4}]] <>
  " (R^2=" <> ToString[Round[r2Decay, 0.01]] <> ") ", "N/A"]];

Print[Grid[shollTable, Frame → All, Background → {None, {LightGray, None}},
  Alignment → Left, ItemSize → {{22, 28, 15}, Automatic}]];

[1/2] 正在点燃并发暴胀宇宙...
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[2/2] 演化停止，启动全息肖尔形态学分析...



| Sholl Metric (Neuro-Morphology) | Cosmological Physical Mapping           | Extremity Index |
|---------------------------------|-----------------------------------------|-----------------|
| Primary Branches                | Big Bang Initial Degrees of Freedom     | 25              |
| Sholl Critical Radius (r_c)     | Turnaround Time (Expansion to Collapse) | 6               |
| Sholl Critical Value (N_max)    | Maximum Equator Spatial Volume          | 3500            |
| Ramification Index              | Inflation Amplification Factor          | 140             |
| Sholl Regression Coeff (k)      | Big Crunch Topological Deceleration     | -1              |

In[1569]:=

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(* ===== *)
(*5. 3D 全息视界投影 (3D Holographic Horizon Projection)*)
    偏导      偏导      投影
(* ===== *)
Print[Style["正在生成 3D 拓扑视界立体投影...", Purple, Bold]];
    打印      样式      偏导      紫色      粗体

macroUniversePlot3D = Graph3D[universe,
    三维图

    GraphLayout → {"SpringElectricalEmbedding", "RepulsiveForcePower" → -2.0},
    图的布局

    VertexSize → 0.4, VertexStyle → Directive[LightBlue, Specularity[White, 10]],
    顶点大小      顶点样式      指令      浅蓝色      反射度      白色

    EdgeStyle → {_UndirectedEdge → Directive[Cyan, Thick, Opacity[0.6]],
    边的样式      无向边      指令      蓝绿色      粗      不透明度
        _DirectedEdge → Directive[Darker[Gray], Thin, Opacity[0.2]]}, Background → Black,
    有向边      指令      暗黑化      灰色      细      不透明度      背景色      黑色

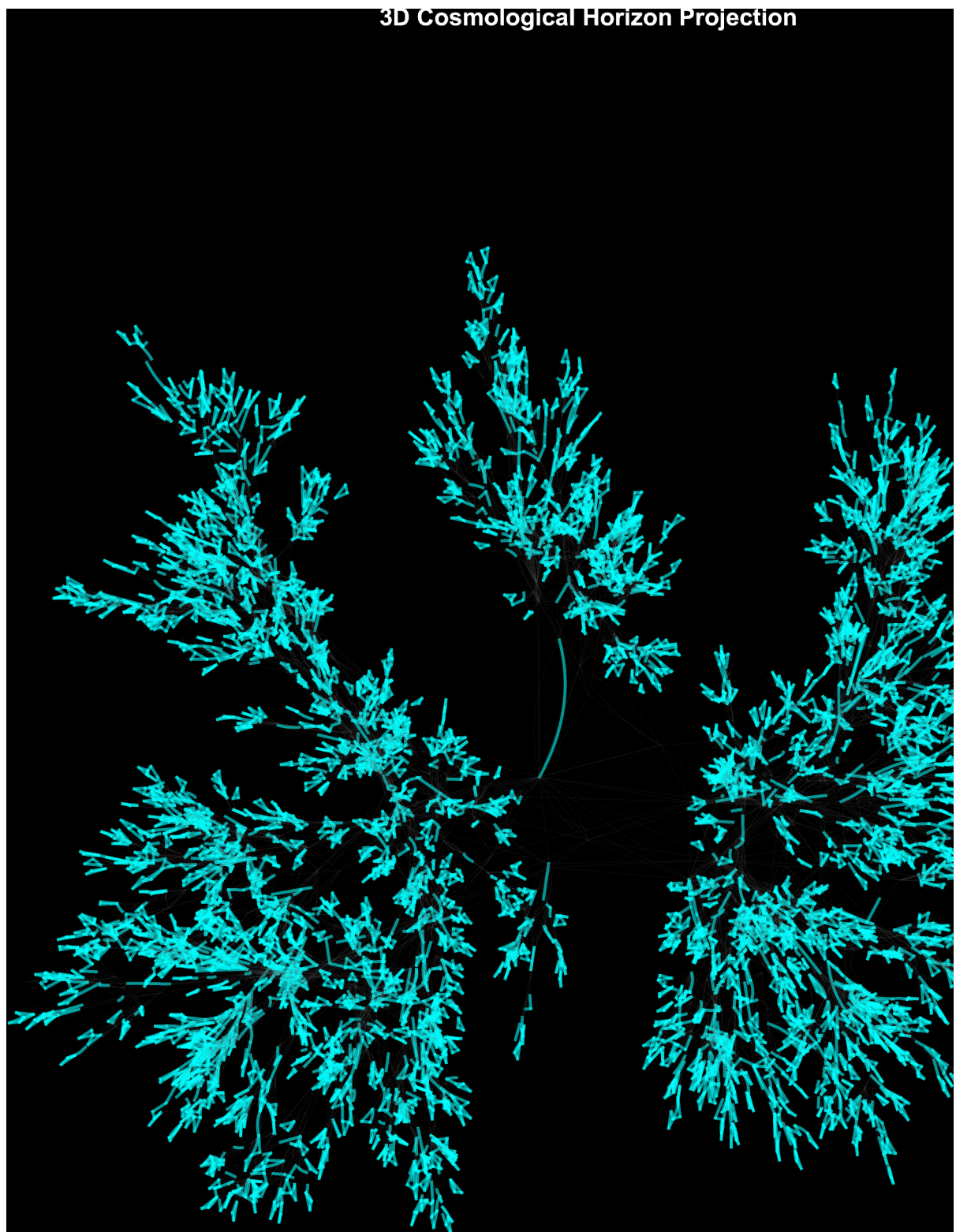
    ImageSize → 800, ViewPoint → {2, -2, 1.5}, (*调整视点以获得最佳几何透视*)
    图像尺寸      视点

    PlotLabel → Style["3D Cosmological Horizon Projection", White, Bold, 16]];
    绘图标签      样式      偏导      投影      白色      粗体

Print[macroUniversePlot3D];
    打印

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正在生成 3D 拓扑视界立体投影...



In[1575]:=

1

Out[1575]=

1